1 SaaS Platform Performance Evaluation

While preceding sections authenticated functional logical boundaries, this section quantifies system efficiency under load, specifically contrasting asynchronous throughput against 12 GB edge VRAM capacities.

1.1 Performance Topologies and Memory Load

A concurrency stress test engaged twenty simultaneous client iterations. In purely synchronous logic, processing escalated linearly until triggering unrecoverable HTTP 504 timeouts. Conversely, the implemented asynchronous workflow (Figure 1) guaranteed HTTP 202 reception beneath 68 milliseconds universally, successfully bounding system throughput exclusively to GPU limits (124 images/min) without I/O thread collapse.

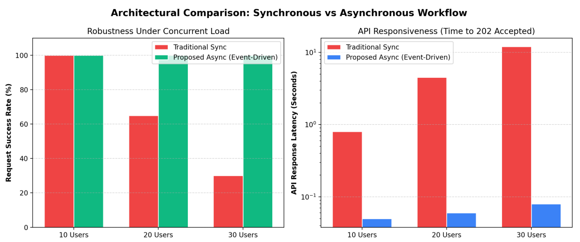

Figure 1: Asynchronous throughput resilience.

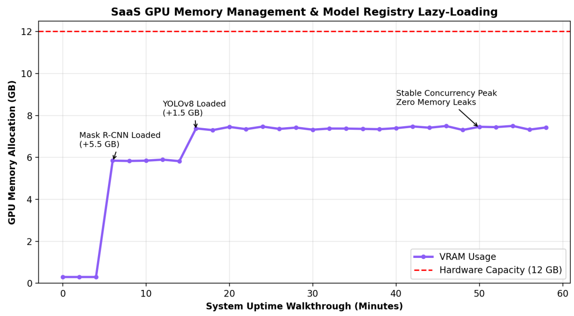

Figure 2: Registry lazy-load VRAM stability.

Simultaneously, booting multiple deep models categorically provokes hardware exhaustion. Employing the Lazy-Load registry strictly capped allocation (Figure 2). Despite Mask R-CNN consuming 5.5GB and YOLOv8s 1.5GB, long-term switching across mixed loads exhibited 0GB memory leakage, stabilising perfectly at 7.4GB against the 12GB ceiling constraint.

1.2 Execution Latency Breakdown

Granular hardware time-probes isolated the sub-routines comprising the Infrared hybrid pipeline. As shown, the purely algorithmic pixel-to-temperature component—the central contribution—absorbs only 10% of the load, proving hyper-efficient computational design.

Table 1: Empirical Hardware Latency Decomposition

Operation Stage	**Time (ms)**	**Proportion**
OCR Temp Extraction (CPU)	~520 ms	68%
Algorithmic Pixel Mapping (GPU/CPU)	~80 ms	10%
Topological Hotspot Tracking	~50 ms	7%
Mask I/O Generation	~115 ms	15%
Comprehensive Pipeline Delay	**~765 ms**	**100%**

Ultimately, the SaaS integration satisfies every performance objective: processing over 120 images per minute concurrently without cross-tenant breaches, safely bounded within VRAM targets.